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Design and Implementation of an Enterprise Data Intelligence Platform for Unified Analytics

Course No.: S1-26_BITSZC426T

Course Title: Dissertation/Project/Project Work

MID SEM PROJECT REPORT

Student Name: DEEKSHA R BHARADWAJ

BITS ID: 2024MC19001

Program: MSC. Information Systems

Project Area: DATABASE SYSTEMS AND APPLICATIONS

Functional Domain: Consumer Goods and Retail Analytics



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(September,2026)

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1. Abstract

Retail and consumer goods organizations generate data across multiple business functions such as sales, inventory, procurement, customer management, products, stores, vendors, and promotions. These datasets may originate from heterogeneous systems and exist in different formats, creating challenges in establishing a consistent analytical environment.

This project focuses on the design and implementation of an enterprise data intelligence platform for unified analytics in the consumer goods and retail domain. The proposed architecture integrates heterogeneous data sources using Azure Data Factory, stores the data in Azure Data Lake Storage Gen2, applies data quality and transformation processes, and provides a cloud-based analytical environment using Snowflake. The platform is further designed to incorporate semantic analytics using Snowflake Cortex AI and provide business-oriented analytical interaction through natural language queries.

The implementation currently focuses on establishing the foundational data engineering pipeline using synthetic retail data. Azure resources, Azure Data Factory pipelines, Azure Data Lake Storage Gen2 containers, Snowflake database and schemas, raw data tables, and customer data transformation workflows have been established. A customer-focused semantic analytics use case is planned as the next stage, using the currently prepared customer dataset to demonstrate business-oriented natural language querying.

The project ultimately aims to evaluate the usefulness of semantic AI-enabled analytics in comparison with conventional business intelligence approaches using metrics such as semantic query accuracy, query response time, dashboard performance, AI response relevance, and usability.

2. Background

The retail and consumer goods industry operates through multiple enterprise applications supporting business functions such as sales, inventory, procurement, supply chain, customer management, and finance.

As organizations expand their digital operations, data is generated across heterogeneous systems and in multiple formats. This creates challenges in maintaining consistent, accessible, and reliable analytical data.

A centralized enterprise data intelligence platform can provide a common analytical foundation by integrating heterogeneous sources, applying data quality and transformation processes, and making business-ready data available for analytics.

The proposed project applies this concept to a representative retail environment based on synthetic data. The implementation uses cloud technologies including Azure Data Factory, Azure Data Lake Storage Gen2, Snowflake, Snowflake Cortex AI, and Power BI.

The project is designed not only as a data integration solution but also as an analytical platform in which a semantic layer can provide business-friendly access to enterprise data.

3. Requirement / Problem Statement

Enterprise retail organizations often have business data distributed across multiple operational systems and data formats. Such fragmentation can result in inconsistent reporting, limited cross-functional visibility, and additional effort for business users attempting to obtain analytical information.

Traditional business intelligence systems provide structured reporting and dashboards but commonly rely on predefined reports, technical users, or knowledge of the underlying data structure.

Therefore, this project investigates an enterprise data intelligence architecture that combines:

- heterogeneous data integration,
- centralized cloud storage,
- data quality and transformation,
- cloud data warehousing,
- semantic analytics,
- natural-language interaction,
- AI-assisted analysis, and
- business intelligence dashboards.

The research component focuses on evaluating whether a semantic AI-enabled analytical layer can support business-user self-service analytics with respect to query accuracy, usability, response relevance, and decision-making efficiency.

The approved project outline identifies these same evaluation dimensions and objectives.

4. Motivation

The project is motivated by the increasing adoption of cloud-based data platforms and AI-assisted analytics.

From a technical perspective, the project provides an opportunity to implement an end-to-end data engineering architecture involving:

- cloud data integration,
- data lake storage,
- data transformation,
- data quality validation,
- data warehousing,
- incremental processing,
- semantic analytics, and
- business intelligence.

From a research perspective, the project provides an opportunity to investigate how semantic AI can change the way business users interact with enterprise analytical data.

The use of synthetic retail data also enables the architecture to be demonstrated without using confidential organizational or client information.

5. Project Objectives

The project objectives are:

- Design a scalable enterprise data intelligence architecture.
- Integrate heterogeneous retail data sources.
- Establish a centralized analytical data environment.
- Implement data validation and transformation processes.
- Develop a layered analytical data model.
- Implement incremental data processing where applicable.
- Develop a semantic analytics layer using Snowflake Cortex AI.
- Enable natural-language interaction with business data.
- Evaluate semantic query accuracy and response relevance.
- Measure system and analytical performance.
- Develop business intelligence dashboards using Power BI.
- Compare semantic AI-enabled analytics with conventional BI workflows.

These objectives are consistent with the approved project outline.

6. Project Scope

The project includes:

- Synthetic retail data
- Multiple data formats
- Azure Data Factory
- Azure Data Lake Storage Gen2
- Data quality validation
- Data cleansing
- Duplicate handling
- Incremental loading
- Snowflake data warehouse
- Raw, staging, and curated layers
- Metadata and lineage considerations
- Semantic analytics
- Snowflake Cortex AI
- AI Analytics Agent
- Power BI dashboards
- Performance evaluation
- Cost optimization considerations
- Conceptual security controls

7. Data Source Design

The project uses representative synthetic retail data to simulate heterogeneous enterprise sources.

Source Format	Representative Entities	Status
REST API	JSON / Sales / incremental retail	data Prepared
CSV	CSV Customers, Sales, Products, Stores, Vendors, etc.	Implemented
Parquet	Analytical/incremental datasets	Prepared/planned

The original project outline identifies REST APIs, Excel, CSV, Parquet, and relational database tables as the planned heterogeneous sources.

8. Retail Dataset

The current synthetic retail dataset contains the following major entities:

- Customers
- Products
- Sales
- Inventory
- Promotions
- Purchase Orders
- Stores
- Vendors

The dataset was prepared to represent a supermarket/retail business environment without using confidential client data.

9. Azure Environment Implemented

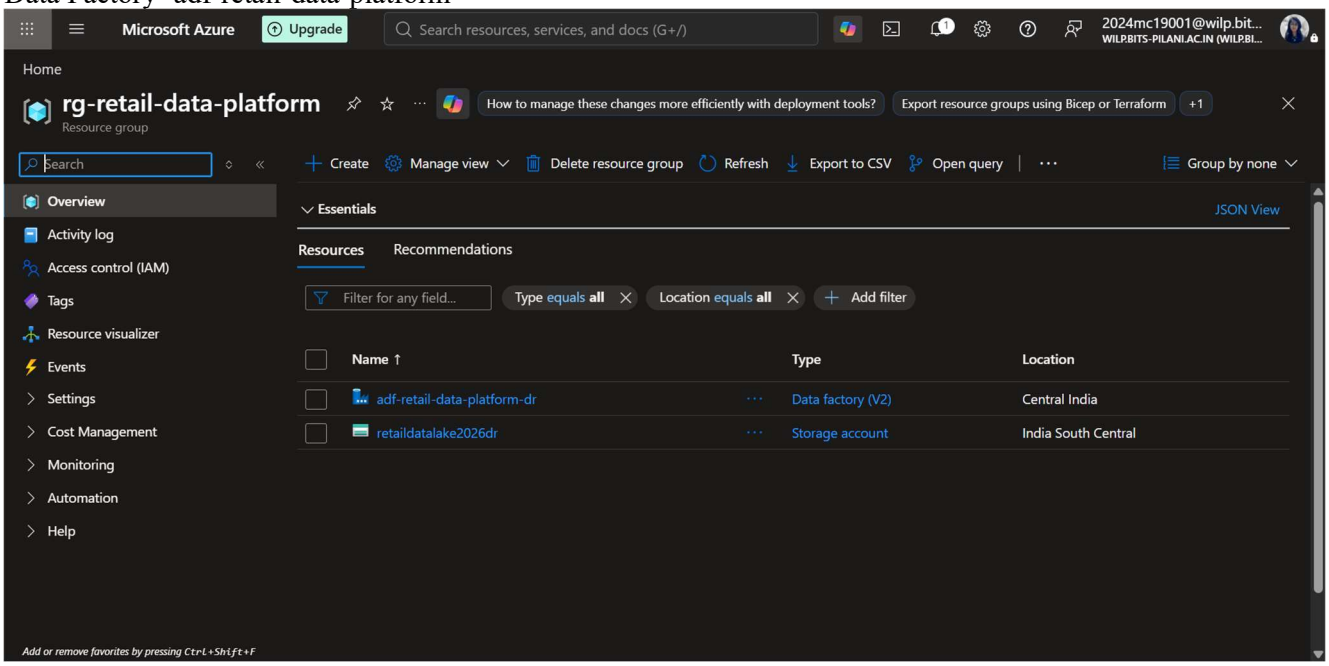
The following Azure resources have been established.

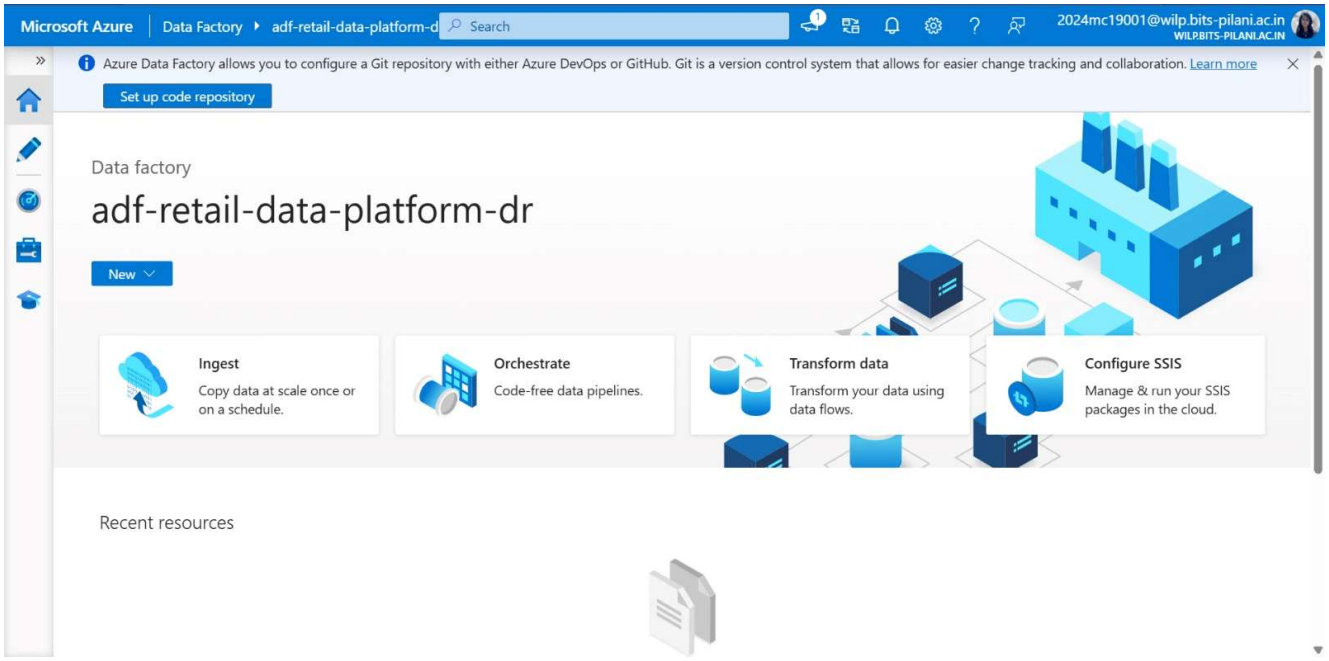
- Resource Group
- rg-retail-data-platform
- Storage Account
- retaildatalake2026dr
- Data Lake Containers
- raw
- staging
- curated
- adf-staging

10. Azure Data Factory Implementation

Azure Data Factory has been configured as the primary orchestration and data integration service.

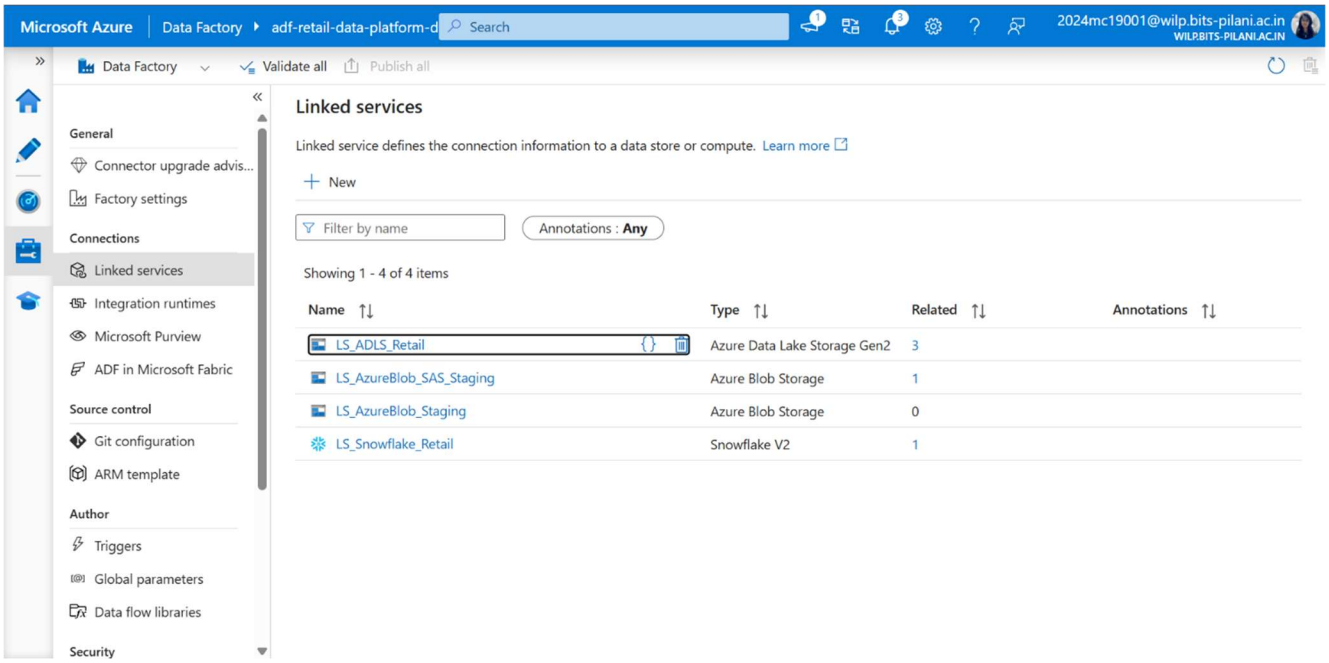
Data Factory- adf-retail-data-platform





Linked Services

- LS_ADLS_Retail
- LS_Snowflake_Retail
- LS_AzureBlob_SAS_Staging



The ADLS linked service provides access to the Azure Data Lake environment.

The Snowflake linked service connects ADF with the Snowflake analytical environment.

A temporary Blob staging configuration is also being used for the ADF-to-Snowflake staged-copy operation.

Process to Load Data into Raw Layer using Azure Data Factory

- ✓ Connect Data Sources – Configure ADF linked services and datasets for sources such as CSV, Excel, Parquet, REST APIs, and relational databases.
- ✓ Extract Data – Use ADF pipelines and Copy Activity to extract data from the configured source systems.
- ✓ Store in Raw Layer – Load the extracted data into the Raw layer in Azure Data Lake Storage Gen2 (ADLS) while preserving the original source data.
- ✓ Apply Incremental Loading – Use a timestamp/watermark or CDC mechanism to identify and load only new or modified records during subsequent pipeline runs.
- ✓ Validate and Monitor – Validate file/record counts and schema consistency, and monitor pipeline execution, failures, and data movement through ADF monitoring.

11.Raw Data Loading

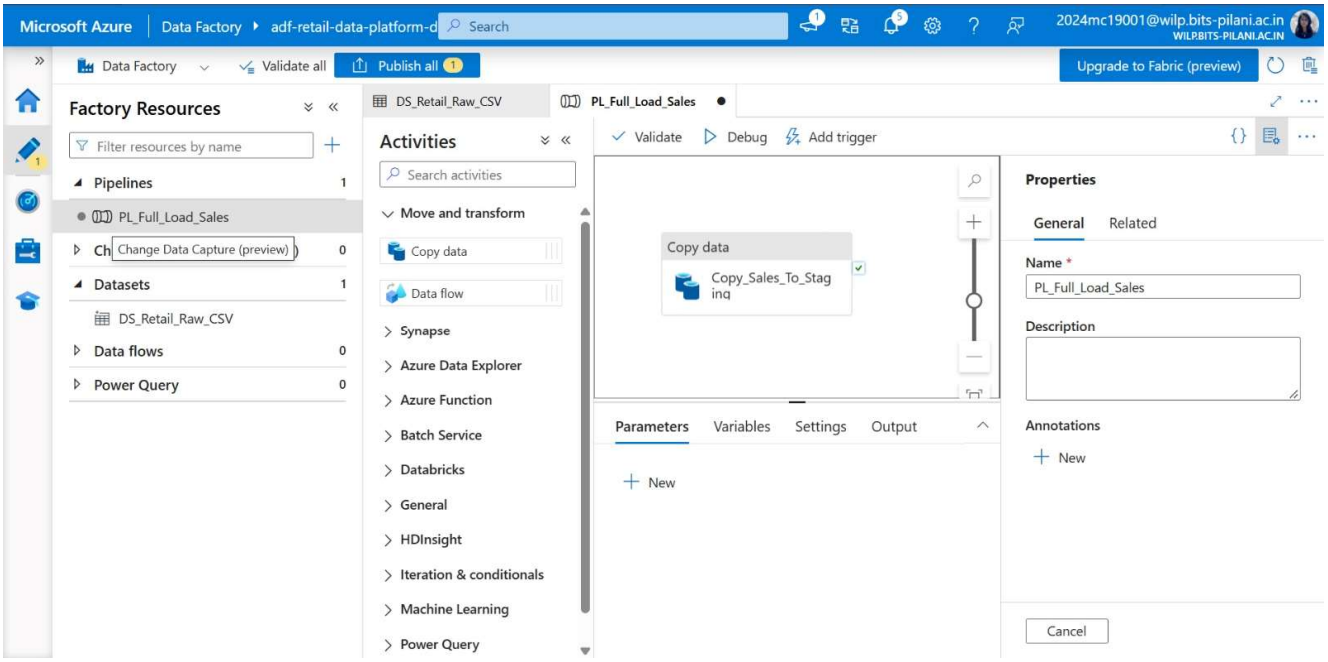
The synthetic retail CSV files have been uploaded into the corresponding raw folders in ADLS Gen2.

For example:

raw/customers/customers.csv
raw/sales/sales.csv
raw/products/products.csv
raw/inventory/inventory.csv

ADF datasets have been configured to access the raw files.

A sales staging pipeline was also created to demonstrate movement of raw data into an intermediate staging location.



12. Customer Data Cleaning

The customer data transformation includes standardization and validation.

Customer Name
`trim(Customer_Name)` – This removes unnecessary leading and trailing whitespace.

Gender
`upper(trim(Gender))` – This standardizes gender values.

City
`initCap(trim(City))` – This standardizes city representation.

Customer Segment
initCap(trim(Customer_Segment)) – This standardizes customer-segment values

13. Customer Data Validation

The current validation rule ensures that required customer attributes are present and that customer age falls within the defined range.

```
!isNull(Customer_ID)
&& !isNull(Customer_Name)
&& !isNull(Age)
&& Age >= 18
&& Age <= 100
```

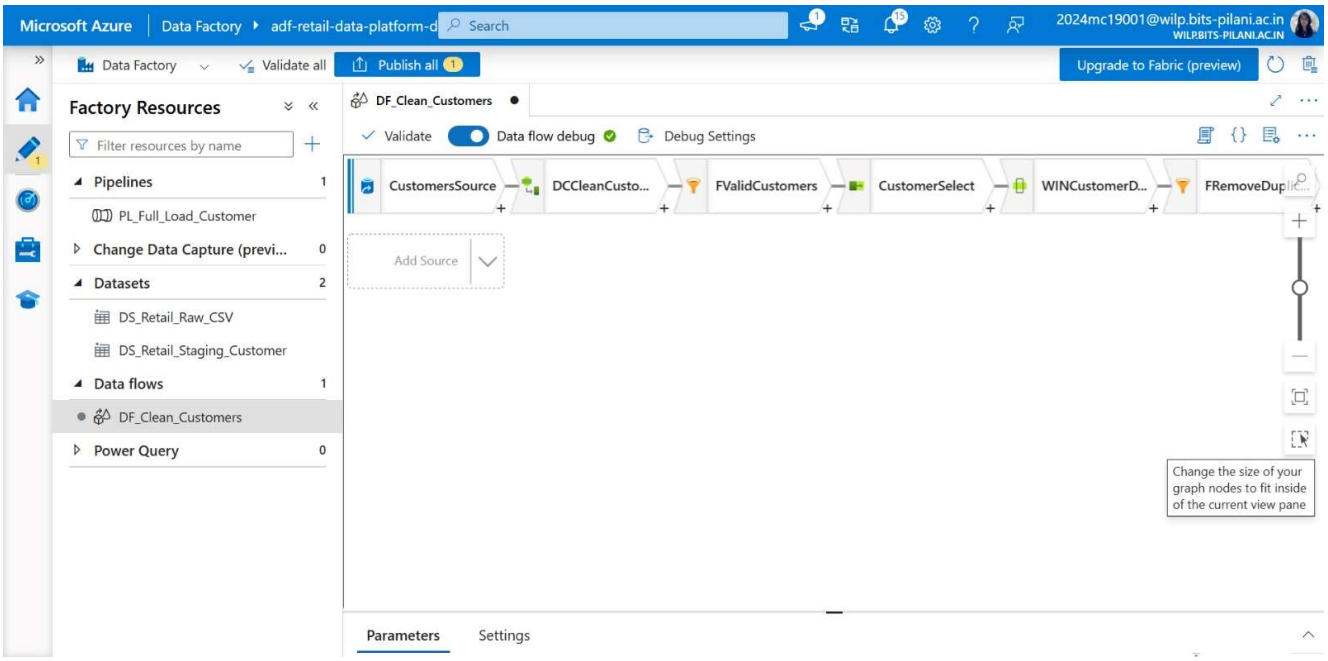
This demonstrates business-rule validation in the transformation layer.

14. Duplicate Handling

Duplicate customer records are handled using a Window transformation.

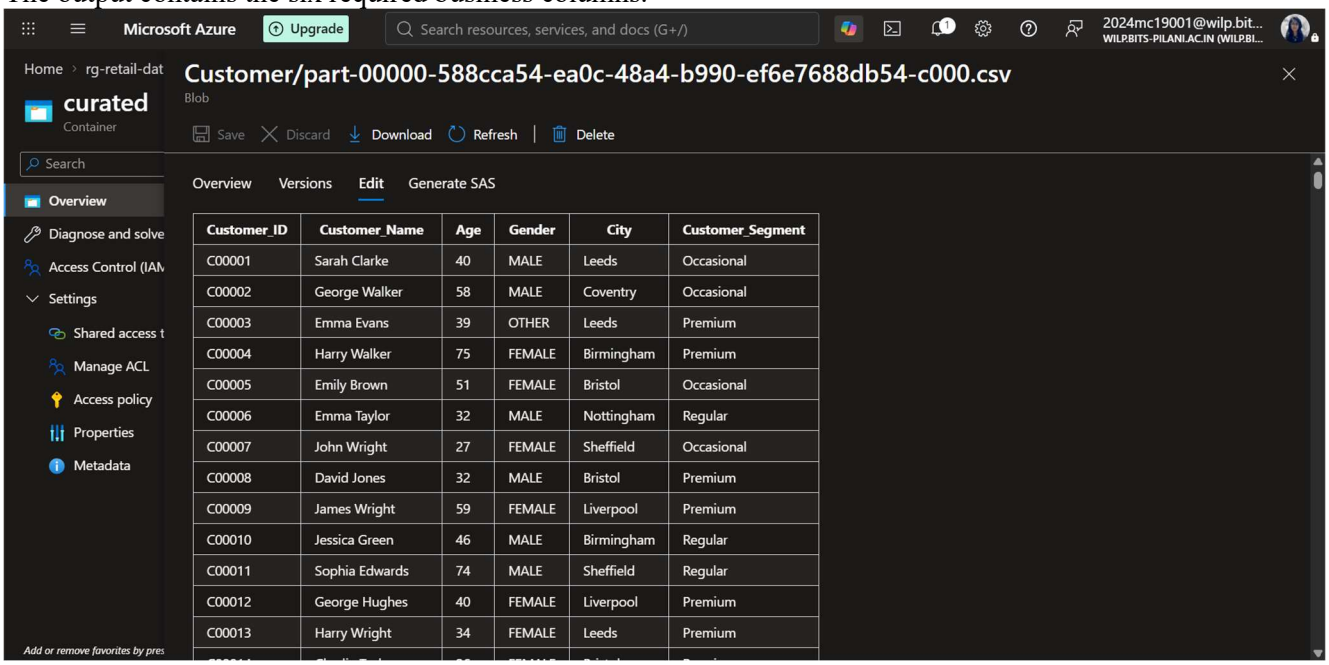
The logic uses:
`rowNumber()`
with:
Partition By:
Customer_ID
and:
Order By:
Customer_ID ASC
The first record is retained:
`row_num == 1`

The temporary row_num field is removed before the final sink.
This demonstrates duplicate detection and removal as part of the data quality process.



15. Curated Customer Data

The cleaned customer dataset has been successfully written to the curated layer.
The output contains the six required business columns:



Customer_ID	Customer_Name	Age	Gender	City	Customer_Segment
C00001	Sarah Clarke	40	MALE	Leeds	Occasional
C00002	George Walker	58	MALE	Coventry	Occasional
C00003	Emma Evans	39	OTHER	Leeds	Premium
C00004	Harry Walker	75	FEMALE	Birmingham	Premium
C00005	Emily Brown	51	FEMALE	Bristol	Occasional
C00006	Emma Taylor	32	MALE	Nottingham	Regular
C00007	John Wright	27	FEMALE	Sheffield	Occasional
C00008	David Jones	32	MALE	Bristol	Premium
C00009	James Wright	59	FEMALE	Liverpool	Premium
C00010	Jessica Green	46	MALE	Birmingham	Regular
C00011	Sophia Edwards	74	MALE	Sheffield	Regular
C00012	George Hughes	40	FEMALE	Liverpool	Premium
C00013	Harry Wright	34	FEMALE	Leeds	Premium

The generated part-...csv output is a normal result of the Spark-based Mapping Data Flow execution.

16. Snowflake Environment

A Snowflake database has been established for the analytical platform.

Database- RETAIL_DB
Schemas-
RETAIL_DB.RAW
RETAIL_DB.STAGING
RETAIL_DB.CURATED

The initial customer table has been created in the RAW schema.

```
CREATE OR REPLACE TABLE CUSTOMERS (
    CUSTOMER_ID VARCHAR,
    CUSTOMER_NAME VARCHAR,
    AGE INTEGER,
    GENDER VARCHAR,
    CITY VARCHAR,
    CUSTOMER_SEGMENT VARCHAR
);
```

17. Snowflake Integration Status

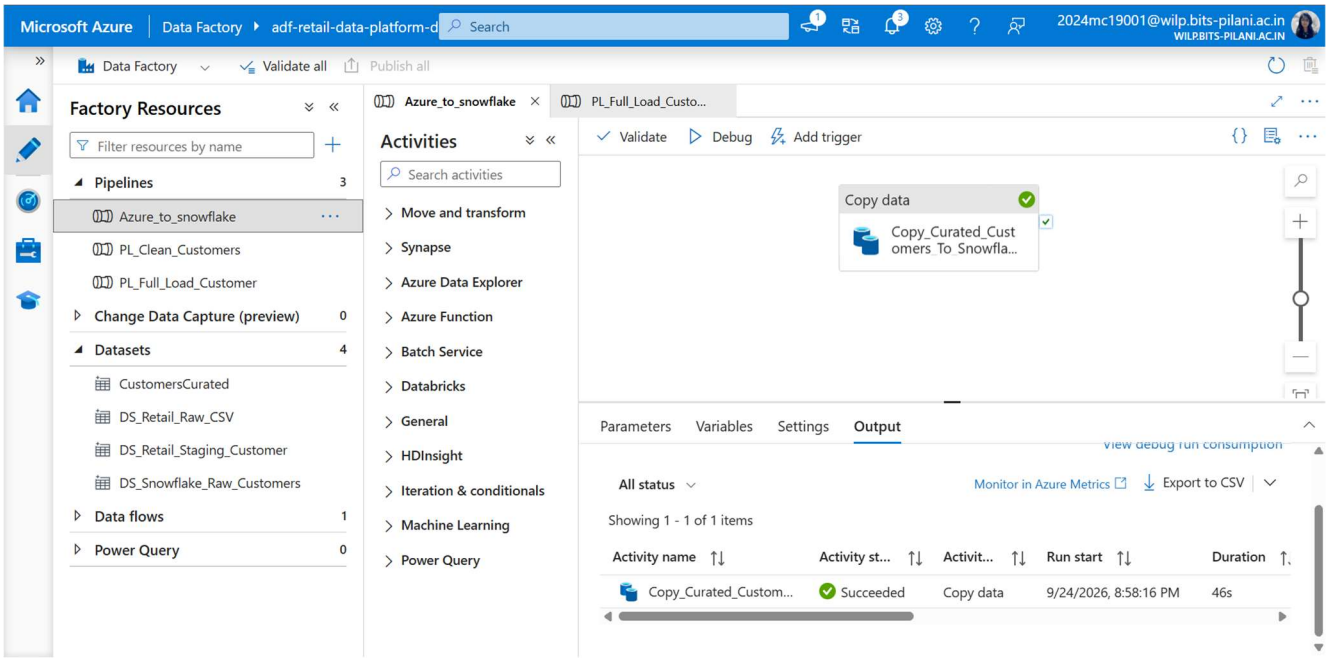
An ADF Snowflake linked service has been configured using: LS_Snowflake_Retail

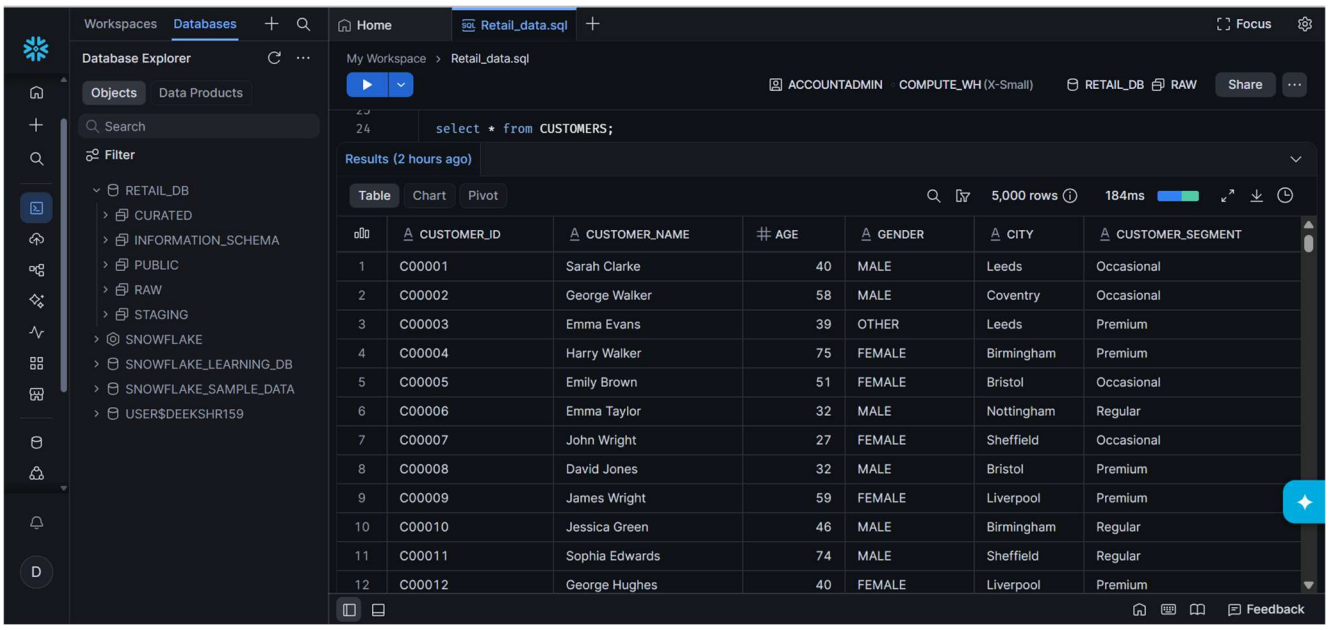
The Snowflake account configuration was corrected and the connection progressed to the database/table stage.

A pipeline was created: PL_Load_Customers_Snowflake

With the copy activity: Copy_Curated_Customers_To_Snowflake

The source is configured to read: curated/Customer/*.csv and the destination is the Snowflake customer table.





18. Initial Semantic Analytics Use Case

The next implementation stage is to develop a customer-focused semantic model using the customer data already prepared.

The semantic model will initially use the available attributes:

Customer_ID
Customer_Name
Age
Gender
City
Customer_Segment

The semantic model will provide business definitions and relationships that allow natural-language questions to be translated into appropriate analytical queries.

19. Initial Semantic Queries

The first semantic model will support questions such as:

Customer Count- How many customers are there?
Expected logic: COUNT(Customer_ID)

Customer Segmentation- How many customers are there in each customer segment?
Expected logic: GROUP BY Customer_Segment

Gender Distribution- What is the distribution of customers by gender?
Expected logic: GROUP BY Gender

City Distribution- How many customers are there in each city?
Expected logic: GROUP BY City

Average Age- What is the average age of customers?
Expected logic: AVG(Age)

Segment Age Analysis- What is the average age of customers in each customer segment?
Expected logic:
GROUP BY Customer_Segment
AVG(Age)
City and Segment

How are customers distributed across segments in different cities?
Expected logic: City+ Customer_Segment+ COUNT(Customer_ID)

These queries provide the initial basis for evaluating natural-language semantic interpretation.

20. Semantic Query Design

For every semantic query, the project will document four components:

- Business Question
- Semantic Definitions
- Generated Analytical Query
- Business Result

For example: Business Question

How many customers belong to each customer segment?

Semantic interpretation:
Customer = CUSTOMER table

Customer segment= CUSTOMER_SEGMENT

Number of customers= COUNT(CUSTOMER_ID)

Grouping= CUSTOMER_SEGMENT

This provides a concrete way to evaluate whether the semantic layer correctly interprets business terminology.

21. Planned Semantic Expansion

After the customer semantic model is validated, the same approach can be expanded to other retail entities:

- Customer
- Product
- Sales
- Inventory
- Store
- Vendor
- Purchase Order
- Promotion

Eventually, cross-entity questions can be supported once the corresponding Snowflake tables and relationships have been implemented.

For example:
Which products generated the highest sales?

Which stores have the highest sales?

Which products are currently low in inventory?

Which customer segments generate the highest revenue?

These should only be implemented after the required datasets and relationships are available

22. Incremental Data Processing

The project also includes an incremental retail dataset and an incremental REST API package.

The objective is to process newly created or modified records instead of repeatedly processing the complete dataset.

The original methodology also specifies timestamp/watermark-based incremental loading and evaluation of CDC where feasible.

23. Data Quality Framework

The project incorporates the following data-quality checks:

Check	Purpose
Null validation	Identify missing required values
Duplicate detection	Prevent duplicate business records
Schema validation	Ensure expected columns and types
Data-type validation	Maintain consistent data types
Range validation	Identify invalid numerical values
Business-rule validation	Enforce domain-specific rules
Standardization	Maintain consistent text representation

The implemented customer flow already demonstrates null validation, age-range validation, text standardization, and duplicate handling.

24. Layered Data Architecture

The platform uses three logical data layers.

Raw
Purpose: Contains source-oriented data with minimal modification

- Traceability
- Reprocessing
- Historical preservation
- Source representation
- Staging

ADLS Raw
Purpose: Used for intermediate transformation and validation.

- Cleansing
- Validation
- Standardization
- Duplicate handling
- Intermediate processing

Curated- Contains business-ready analytical data.
Purpose: The same layered concept is also reflected in the planned Snowflake architecture.

- Reporting
- Semantic analytics
- Business intelligence
- Analytical querying

The same layered concept is also reflected in the Executed Snowflake architecture.

25. Performance and Cost Optimization

The project will evaluate performance optimization techniques including:

- Query pruning
- Clustering
- Appropriate warehouse sizing
- Materialized views where applicable
- Efficient data processing
- Incremental loading

Cost optimization considerations include:

- Auto-suspend
- Auto-resume
- Appropriate compute sizing
- Avoiding unnecessary full reloads
- Storage optimization

These areas are included in the original approved methodology.

26. Security Considerations

Full enterprise security implementation is outside the project scope.

However, the architecture conceptually considers:

- Role-Based Access Control
- Role hierarchy
- Least-privilege access
- Row-level security
- Data masking

The project uses synthetic data, avoiding exposure of confidential organizational or client information.

27. Risks and Mitigation

Risk	Mitigation
Heterogeneous data formats	Standardized ingestion and transformation using ADF
Data quality issues	Validation and cleansing during transformation
Duplicate records	Duplicate detection and record filtering
Integration complexity	Modular layered architecture
Snowflake connectivity issues	Separate linked-service testing and staged integration
Semantic query inaccuracies	Explicit semantic definitions and business rules
AI response inaccuracies	Query validation against expected results
Cloud cost	Auto-suspend, appropriate compute sizing and controlled execution
Schedule delays	Prioritize core pipeline and customer semantic use case before expansion

28. Conclusion

The project has progressed from architectural planning to implementation of the foundational enterprise data pipeline.

A synthetic retail dataset has been established and integrated into Azure Data Lake Storage Gen2. Azure Data Factory has been configured for data movement and transformation, and a customer-specific data quality and transformation workflow has been successfully implemented. The customer pipeline performs standardization, validation, and duplicate handling before producing curated analytical data.

A Snowflake analytical environment has also been established with separate Raw, Staging, and Curated schemas, and the initial customer table has been created. The final curated-to-Snowflake loading step is currently being completed.

The next major milestone is the implementation of a customer-focused semantic model using the curated customer data. This will provide the first practical demonstration of the project's semantic AI objective through natural-language business questions and measurable query results.

The subsequent stages will expand the semantic model to additional retail entities, introduce the AI Analytics Agent, develop Power BI dashboards, and evaluate the complete platform using the predefined performance and usability metrics.